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RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY

Bachelor of Information and Communication Technology
First Year - Semester II Examination – October 2020

ICT 1305 – PROGRAM DESIGN AND PROGRAMMING

Time: Two (02) hours

Answer any **Four (04)** questions in the answer booklet.

1.

a) Explain the differences between 1D and 2D arrays using suitable examples. **[05 marks]**

b) Briefly explain what is an array and advantages of using arrays. **[05 marks]**

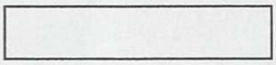
c) Write a program in C to obtain three test scores from the user and store them in an array. Print the scores with average value. **[07 marks]**

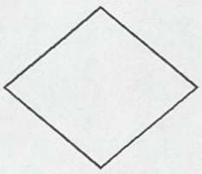
d) Draw a flowchart for finding the largest value between three user given numbers. **[08 marks]**

[Total 25 marks]

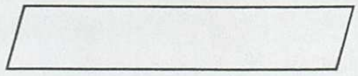
2.

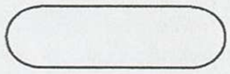
a) Identify the given Flowchart Symbols. Give each one a suitable example.

i. 

ii. 

iii. 

iv. 

v. 

[5x02 marks]

b) Draw a flowchart for the algorithm given below.

Step 01: regular_pay = gross_pay = ot_pay = 0
Step 02: GET pay_rate and hours_worked
Step 03: IF hours_worked <=40 then
    *regular_pay = hours_worked * pay_rate*
  ELSE
    *regular_pay = 40 * pay_rate*
    *ot_pay = (hours_worked – 40) * 1.5 * pay_rate*
Step 04: gross_pay = regular_pay + ot_pay
Step 05: PRINT gross_pay

[07 marks]

c) Identify three modes to open files in C program. **[03 marks]**

d) Complete the following function using pointers to swap two given numbers.

```
______ swap (int _____, int *py)
{
        int _______;
        temp = *px;
        _____ = *py;
        *py = _______;
}
```

[05 marks]

[Total 25 marks]

3.

a) List five (05) rules in writing pseudocode. **[05 marks]**

b) Briefly explain the given verbs used in pseudocode.

i. READ
ii. PROMPT
iii. DISPLAY

iv. GET

v. WRITE

[05 marks]

c) Convert the given code into the relevant pseudocode. **[07 marks]**

```
#include <stdio.h>
int main()
{
        int rows;
        printf("Enter number of rows: ");
        scanf("d%", &rows);
        for(int i = 0; i<=rows; i++)
        {
                for(int j =0; j<=i; j++);
                {
                        printf("*");
                }
                printf("\n");
        }
        return 0;
}
```

d) Write a program in C to print individual characters of a string in reverse order.

[08 marks]

[Total 25 marks]

4.

a) Differentiate user defined functions and standard library functions. **[05 marks]**

b) What is a recursive function? State the advantages of using recursion over repetition. **[05 marks]**

c) Complete the given function to output the power of a given number.

```
#include <stdio.h>
int power(int,int);
void main()
{
        int n,m,k;
```

```
        printf("Enter the values of n and m:");
        scanf("%d%d",&n, &m);
        k = power(n,m);
        printf("The value of n*m for n= %d and m=%d is: %d",n,m,k);
}
int power(int x, int y)
{

}
```

[05 marks]

d) Write an algorithm to input a number n and print the numbers from 1 to n. Modify and rewrite the algorithm to add the multiples of three and five between 1 and n, only if the entered number is a multiples of three or five.
(ex: if n=10, 3+5+6+9 = 23) **[05 marks]**

[Total 25 marks]

5.

a) List five properties of a good algorithm. **[05 marks]**

b) What are the advantages of using flowcharts to represent algorithms? **[05 marks]**

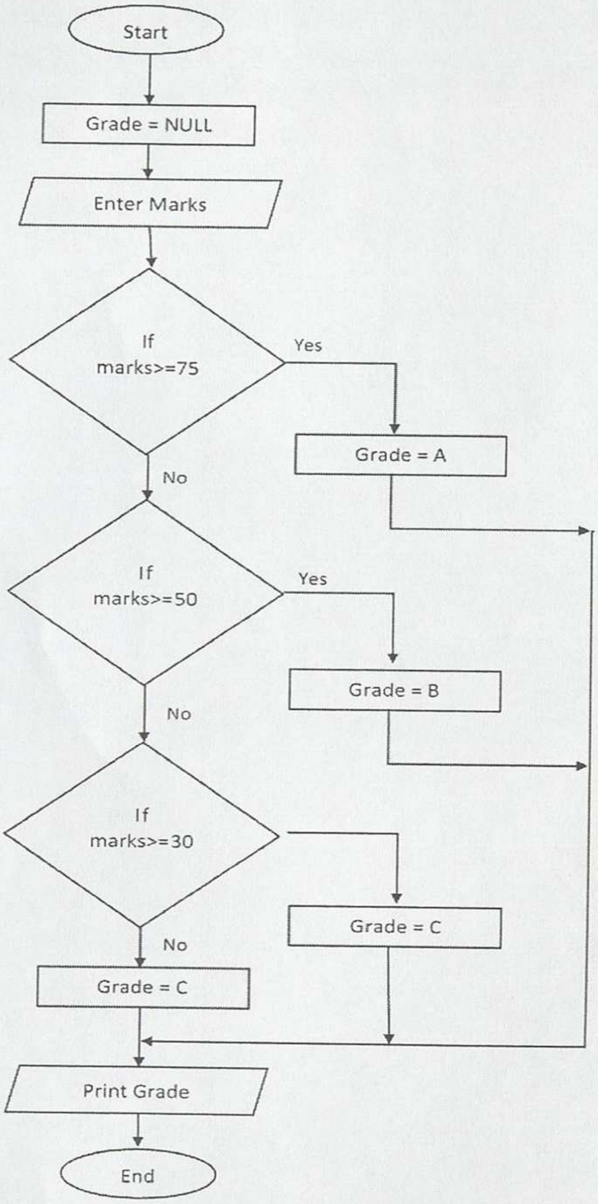
c) Create a Structure in C for a Book which store Title, Author and Book_ID. Create three variables and save the following books in them.

<u>Book 1</u>
Title: To Kill a Mockingbird
Author: Harper Lee
Book_ID: 001

<u>Book 2</u>
Title: Little Women
Author: Lousia M.Alcott
Book_ID: 002

<u>Book 3</u>
Title: Lord of the Flies
Author: William Golding
Book_ID: 003 **[08 marks]**

d) Convert the given flowchart into a relevant algorithm.



[07 marks]

[Total 25 marks]

--End--